KNECK OFFICIAL

Computer Studies — Paper 3

451/3 • ORIGINAL REVISION SET 01 • Suggested 6–8 weeks (revision project) • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

This is a PROJECT component, not a timed written paper. Complete the brief, submit the working system and evidence, and demonstrate it to a teacher. The official 2026 timetable assigns project components January–July; this shorter original exercise is for revision only.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

PROJECT BRIEF — LIBRARY RESOURCE CATALOGUE

Complete all deliverables.

1. Analyse the needs of a school library that wants to catalogue books and record borrowing by staff-administered entries. Interview a teacher or use a clearly labelled fictional scenario. Submit problem statement, three measurable objectives, scope, requirements and a realistic work plan. [15 marks]

2. Design a solution with books, borrowers and loans. Submit data dictionary with keys/types/validation, entity relationship diagram, at least three screen designs and pseudocode for issue/return with availability checking. [20 marks]

3. Build a working prototype using an accessible tool of your choice. It must add/edit/search books, issue/return an item, prevent negative availability, validate input and produce an overdue report using an explicit reference date. Include a backup/export function. [35 marks]

4. Test at least ten normal, boundary and invalid cases with expected/actual results. Fix defects and retest. Include evidence of restoring an exported backup. [15 marks]

5. Submit technical and user documentation, a short demonstration, evaluation against objectives and acknowledgement of reused libraries or tutorials. [15 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

PROJECT BRIEF — LIBRARY RESOURCE CATALOGUE

1. (15 marks) Problem/context3; measurable objectives3; functional/non-functional requirements4; scope/privacy2; schedule/resources3. Do not collect real student personal data for a revision prototype.

2. (20 marks) Data dictionary6; relationships4; screens4; issue/return logic4; validation/security design2.

3. (35 marks) Book CRUD/search8; issue/return8; availability/integrity6; validation4; overdue report5; backup/export4. Demonstrate with fictional records and clearly documented run steps.

4. (15 marks) Diverse planned cases5; expected/actual evidence5; defect fixes/retest3; restore evidence2.

5. (15 marks) Reproducible setup/user guide5; demonstration4; honest limitations/evaluation4; attribution2. Teacher asks live questions to establish understanding and authorship.

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf